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SGI IrisVision

[@8EE6.ADF](#) SGI Micro-Channel IRISVISION Adapter

[The Very Unofficial IrisVison Home Page](#) by Roger Brown, local copy (original archived [HERE](#))

[SGI IrisVision add-in 3D accelerator for PC \(1990\)](#) (by Jiří Zima)

[Graphics Library Programming Guide, Volume I](#)

[Graphics Library Programming Guide, Volume II](#)

[IrisGL Application Programming Interface](#) (same as above, but HTML, archived)

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RS/6000

Known as "sabine" or .devices.mca.8ee6 in the [RS/6000 world](#). Two variants:

"FC (2780) 8-Bit 3D Color Graphics Processor (Type 1-3)"

w/ 8-bit MDE1 / 71F1117 daughter card ([description](#))

"FC (2781) 24-Bit 3D Color Graphics Processor (Type 1-3)"
w/ 24-bit [MEV2 / 71F1114](#) daughter card ([description](#))

Drivers & Software

IrisVision software, 24-bit color demos, and [IrisView DXF file viewer](#)

- [SGI v2.00 disk images](#)
- Pellucid 2.10 (beta) disk images (w/o IrisView) [ivs210.zip](#)
 - Copy of the [GL.386 file](#) for the above

Autodesk [ADI Combined Display/Rendering driver](#)

- Click here to download the [ISA version of the driver](#)
- Click here to download the [MCA version of the driver](#)

Windows [3.x ver 2.30 display driver](#) disk image

IrisVision [GL Software Developer's Kit v 2.10](#) disk image

- You'll need to set a few env. variables, this file has some example settings
- Here's a [link to on-line version for all the IrisGL API](#) (outdated)

Note to PS/2 Micro Channel Users: The Windows display driver (above) are for the ISA IRISVISION card, only. Unfortunately, it seems when Pellucid developed this driver, they did not include support for the MCA version of the board set.

[IV1WinSrc.zip](#) IrisVision Win 3.1 driver source - Thanks to Resman (Glory be unto him!)

Documentation

[IRISVISION Owner's Guide](#)

[IRISVIEW User Guide](#)

[ADI Driver User's Guide](#)

(all from Lorenzo Mollicone)

IrisVision Technical Reference

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(about 700 pages, from Roger Brown)

Product Description

Ed. The IrisVision hardware **lacks hardware support** for certain "high end" IrisGL features, such as texture mapping, alpha blending, accumulation, and stencil buffers!

The following text is stolen from Roger Brown's [The Very Unofficial IrisVison Home Page](#).

Hardware

The IrisVision hardware is nearly identical to the graphics sub-system in the SGI Personal Iris workstation introduced in 1988. The hardware supports a 5th generation geometry processing pipeline, the GE5, an 8 or 24 bit per pixel frame buffer, and a 24-bit per pixel z-buffer for hidden surface removal. The card set implements in hardware, the entire IrisGL graphics Application Programming Interface (API).

Notable differences from the Personal Iris graphics are:

- Uses 256K VRAM instead of 64K VRAM for reduced size
- Has a 3 color cursor instead of 1 color for increased visibility
- Has VGA pass-through capability for PC compatibility

The card has a rich set of video and rendering modes consistent with the Personal Iris.

Origins of the IrisVision

The product came to life originally as an OEM graphics board set based upon the VME bus in the Personal IRIS. Later, IBM approached SGI to develop a Micro Channel Architecture (MCA) version of the card for use in their newly introduced RS/6000 Unix workstation. IBM licensed the MCA card design as well as the IrisGL graphics library from SGI.

In the process of testing the product, it was discovered that an IBM PS/2 Model 70 personal computer running OS/2 could be used to run diagnostics and test programs on the card much easier than using the RS/6000. So a minimal device driver was written for the card and soon IBM was shipping product. (**Ed.** Roger Brown can't find the "minimal OS/2 drivers" and he works for SGI.)

At some point, the light went off in someone's head; "Why don't we sell this board set for use in PCs?". IrisVision was born. Initially, the MCA card was re-designed to offer some features critical for the PC market, including standard 15-pin VGA-style video output and a 15-pin VGA passthrough input connector. The IBM genlock connector was moved to the top of the card, and stereo display signals were also brought out to the VGA passthrough connector. The card occupied 1 32-bit MCA slot and an adjacent 16/32 bit slot. One or two daughter boards provided framebuffer and z-buffer memory.

Work then began on the design of an Industry Standard Architecture (ISA or AT-bus) version of the card. It would occupy 2 16-bit ISA slots and use the identical daughter cards as the MCA

(and IBM) versions of the board set.

Software

The IrisGL API is implemented in a C-language library developed with the Metaware High-C 32-bit C compiler and the PharLap 32-bit DOS-Extender. It was designed to run in a full screen DOS environment. At the time, M/S Windows did not offer a 32-bit programming environment, so the PharLap DOS-Extender technology was the most sophisticated solution available. It features full 32-bit virtual addressing (2 GB application space), virtual memory support, and seamless integration of real and protected mode programs. The MetaWare High-C compiler is ANSI compatible and is just the ticket for compiling Unix source code to run under DOS-Extender on the IrisVision card.

IrisVision General Components

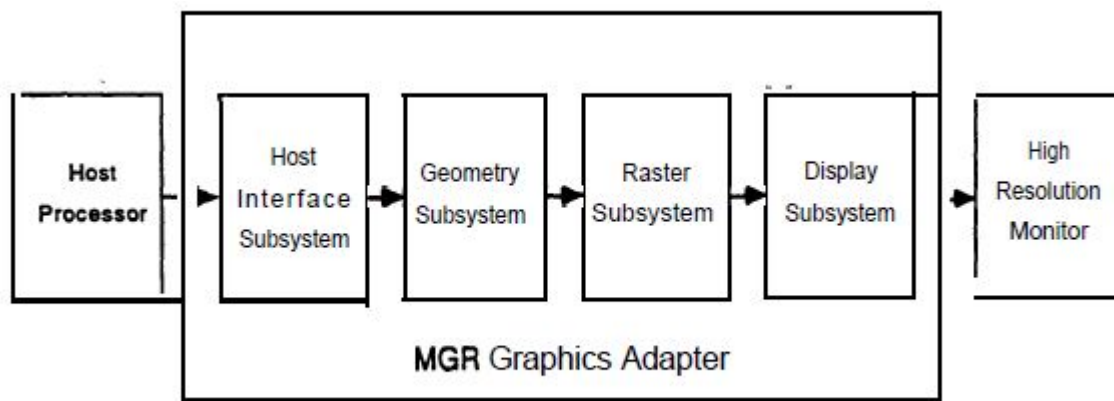


Figure 1.1 MGR Block Diagram

MGE2 Base Card (Card has both GE and HI)

The Host Interface subsystem interfaces to the host processor via the Micro Channel Architecture (MCA) bus. It provides all of the necessary data and control signals required by the MCA bus. It also interfaces to the other three subsystems over a local bus which is compatible with the SGI private bus. The Host Interface Subsystem provides the programmable Option Select (POS) registers which are used to configure the MCA bus interface aspects of the adapter.

MGE2 Geometry Engine Subsystem

The Geometry subsystem contains the Geometry Engine which is used to perform the geometric transformations and lighting calculations on the graphics data. It also performs the clipping and high level rendering calculations before sending the data to the Raster subsystem. The Geometry Engine performs high speed floating point calculations under the control of the onboard microcode.

MGE2 Host System Interface

The host system issues commands to the Geometry Engine by sending command tokens and data parameters down a FIFO. The Geometry Engine reads the FIFO and performs the desired functions. The Geometry Subsystem also contains hardware which controls the addressing of

the hardware components in the Raster and Display Subsystems.

MRV2 Raster Video (both Raster and Display Subsystems)

The Raster subsystem receives instructions and other data parameters from the Geometry Subsystem and does the low level rendering of the data into the image frame buffer bitplanes. It also handles the 2 buffer depth comparisons and updates if the optional Z buffer card is installed. The pixel values for each of the 1.3 million pixels on the 1280 by 1024 high resolution monitor are stored in the video random access memory (VRAM) frame buffer for display on the monitor at a selected refresh rate.

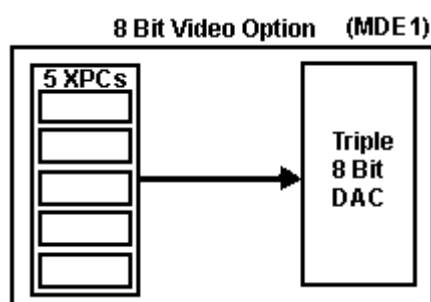
The Raster Subsystem provides the necessary data and control signals to manage the frame buffer bitplanes as well as the Window ID bitplanes and the Auxiliary bitplanes. The Window ID bitplanes are used to control the display format for up to sixteen different on screen windows. The Auxiliary bitplanes are used for drawing pop-up menus, overlays and underlays. The Raster Subsystem also contains two hardware cursor chips.

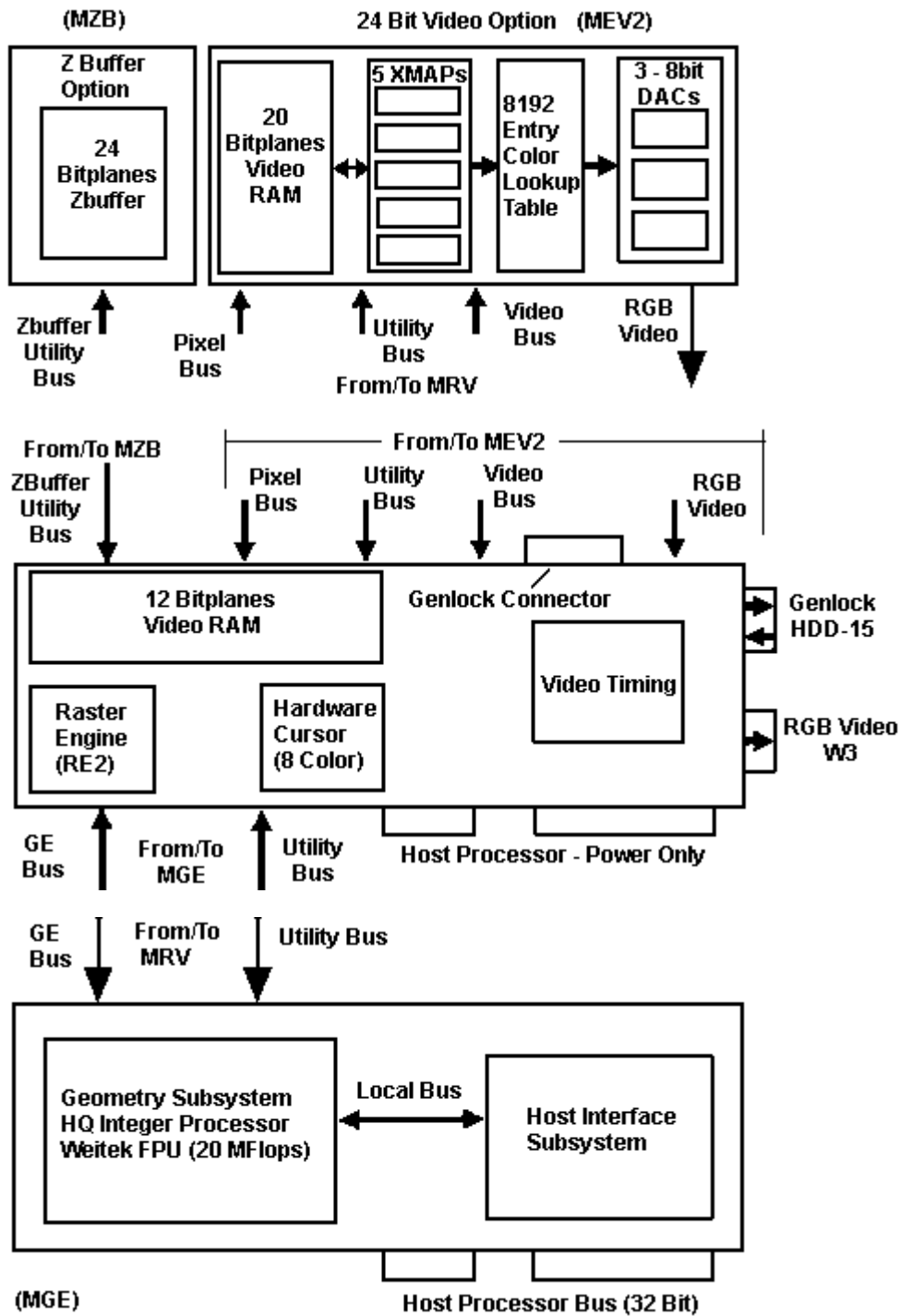
MRV2 Raster Video (both Raster and Display Subsystems)

The Display Subsystem performs the necessary operations to convert the frame buffer image data into analog RGB signals which are sent to the high resolution RGB monitor. The Display Subsystem uses the Window ID bitplane data to determine the format of the pixel data stored in the frame buffer. The frame buffer pixel data can be either a color index value or an RGB value. The frame buffer bitplanes can also be subdivided into two buffers for flicker free motion of on screen objects.

The Display Subsystem performs the necessary operations to access the frame buffer as a single buffer or as a double buffer. The Digital-to-Analog Converters (DACs) are used to convert the digital RGB data into analog RGB data which output to the monitor. The Display Subsystem also provides the necessary timing signals to allow four different types of monitors to be connected to the adapter.

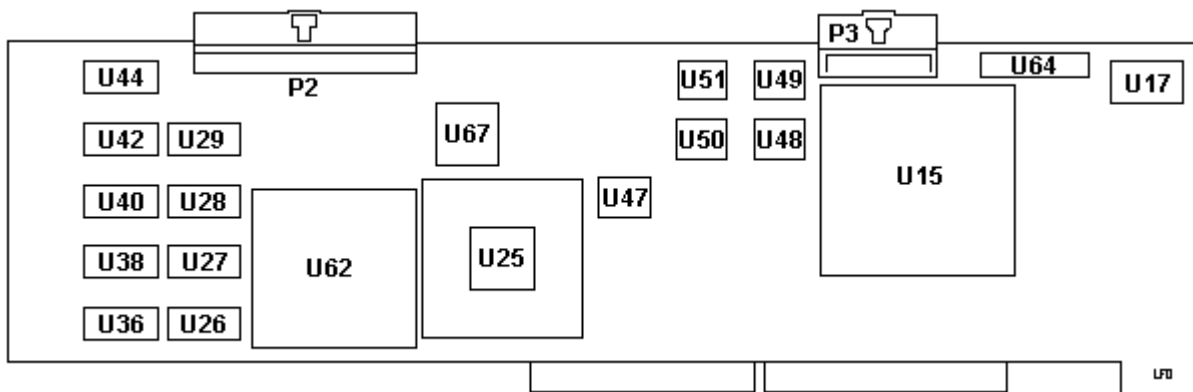
IrisVision Block Diagrams





MGE2 Base Card

FRU P/N 42F6842, P/N 71F1111, OPT. 81F9188, ASSY# 030-8024-003 REV A. 📷 📷 (photos from okqubit.net)



P2 Geometry Engine Bus connector

P3 Utility Bus connector

U15 SGI L1A5205 53F3269

U17 40.000 MHz

U25 SGI L1A3578 HQ1

U26-29,36,38,40,42,44 [IDT7164](#) 8Kx8 SRAM

U47-51 [IDT7201](#) 512x9 FIFO

U62 [XL-3132](#) 100-GCD FPU

U64 MGE 7951 (VPD)

U67 SGI L1A4252 GRF1

This is the Micro Channel Geometry Engine. It is based upon the HQ integer processor from SGI (rated at 10 MIPS) and the Weitek XL-3132 floating point processor (rated at 20 MFLOPS). The geometry engine provides the host interface, handles DMA operations, and handles all the matrix operations, lighting and coordinate transformations.

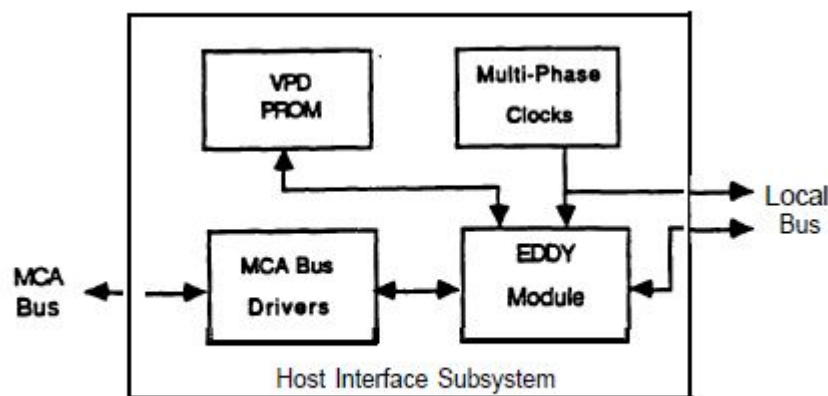


Figure 3.1 Host Interface Subsystem Block Diagram

GRFI Gate Array

Provides two finish flags and corrects some other minor design flaws in the HQL chip.

EDDY Module (U15?)

The Eddy module is the main component of the Host Interface Subsystem. It is responsible for providing the MCA bus interface to the host system. It is also responsible for providing a Local bus interface to the Geometry Subsystem which is compatible with the SGI private bus. The Eddy module is a Bus Master DMA / Memory Slave device for the Micro Channel Architecture. The Eddy module will support a variety of modes of data transfer on the MCA bus, which include:

- Setup Cycles
- 32-bit Memory Slave Cycles

- 32-bit Normal Bus Master Cycles
- 32-bit Bus Master Data Streaming Cycles

The Eddy module only responds to 32-bit Masters when operating as a Memory Slave device. All data transfers to the local bus are 32-bit transfers. The Bus Master DMA feature of the Eddy module will always perform a 32-bit transfer and therefore should always be programmed to transfer data to or from system memory (i.e. 32-bit Slave Memory). The following paragraphs describe the address decoding performed by the Eddy Module.

The Eddy module provides a bus translation from the MCA bus to the Local Bus, the Programmable Option Select (POS) registers and the Control / Status registers

Address Decode Section: responsible for control of read and write operations to POS registers, memory mapped I/O registers and memory mapped control / status registers.

Programmable Option Select (POS) registers contain information to identify and configure the adapter, eliminating the need for hardware jumpers. There are eight directly addressable 8 bit POS registers, as shown in Figure 3.3. Four sub-data registers are accessible through the use of a sub-addressing facility. One of the POS registers indicates channel check status information which is only valid is after an error has caused the Eddy module to Channel Check the host system.

Note: I want to illustrate the options. POS and Bit numbers omitted at random.

- MGR adapter ID
- Interrupt Request (IRQ) Level
- Card enable/disable bit
- Byte Order Selection bit
- Wait State Selection bit
- DMA block transfer size and DMA Bus Master enable/disable bit
- Data Streaming Mode enable/disable bit
- Error checking enable/disable bits .
- Memory address
- Arbitration Level
- Fairness enable/disable bit
- Channel Check Status
- VPD subaddress registers

Note: Channel Check Status / Sub-Address Register (POS 6) has two functions which are controlled by the status bit (bit 6) of the POS 5 register. If the status bit is 1, this register is defined as a sub-address register. If the status bit is a 0, this register contains status information about the pending Channel Check.

Remember the RS/6000 9-K 10/100 bit Ethernet? It has that PCI to MCA bridge chip (ASIC by AMD) and a bog standard PCI Fast Ethernet controller. Speculation is that XPOS are used to

access the PCI Ethernet chip, but we would need to dump the ODT for it in order to find out...

Control / Status Section contains the seven 32-bit registers as shown in Figure 3.4. Some of the 32 bit registers are accessible as multiple 8 or 16 bit registers. The byte ordering of the host machine will affect the address required for accessing the registers when accessing 8 or 16 bit portions of the 32 bit registers. These registers are defined in the registers section of this chapter.

The Eddy module contains two DMA channels. Each channel will arbitrate for the Micro Channel on the same arbitration level. These channels will operate in a serial manner so that a transfer on one channel will complete before a transfer on the other channel can begin. If an error occurs during a DMA transfer, that transfer will be terminated. The other channel will not be able to begin a transfer until the one that was terminated is allowed to complete.

VPD PROM

Only the identification and location of the adapter can be determined from the POS registers. The vital Product Data (VPD) PROM provides additional information about the adapter. It is accessible through the use of the POS Subaddressing Extension which is described later in the section on the POS registers.

The MGR adapter has a VPD PROM on each of the five cards which are used to form the base and enhanced configurations. Each VPD PROM is 256 bytes and contains the following data:

- VPD Header
- VPD ASCII characters
- VPD total length
- CRC on data within the VPD PROM

VPD data fields:

- Card Description
- Engineering Change Level
- Part Number
- Manufacturer
- Field Replaceable Unit Number
- Next VPD Address
- Loadable Microcode Level
- Serial Number - unique for each card

Note: Thank SGI, each VPD chip is socketed into individual pin sockets!

Multi-Phase Clocks Section

Provides the necessary clock signals to the Eddy module. It also provides the various 10 MHz clocks to the Local Bus with the necessary phase relationships.

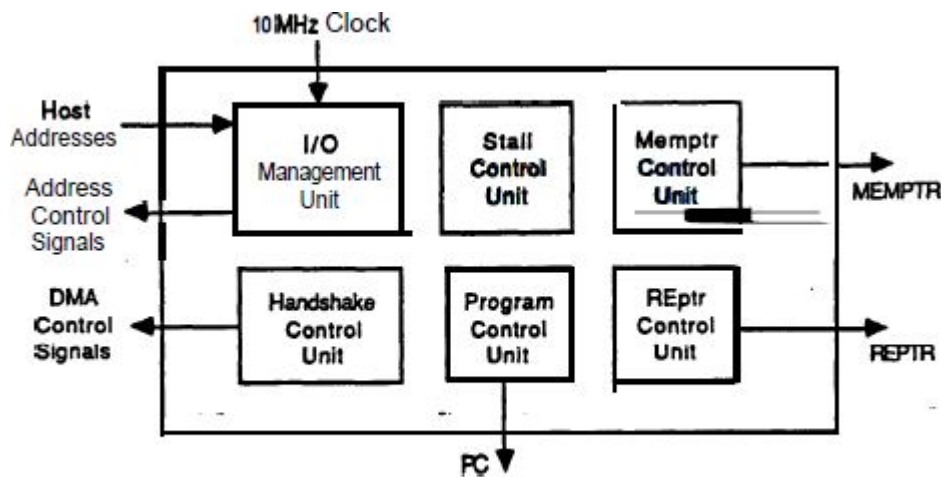


Figure 4.2 HQ1 Functional Block Diagram

I/O Management Unit

Responsible for providing address decoding and bus control for hardware accesses by the Host Interface Subsystem and Geometry Engine.

Handshake Control Unit

Used to provide necessary hardware handshake signals between Host Interface Subsystem, Raster Engine (RE) in the Raster Subsystem, microcode code and data RAM and XMAP2 or XPCI chips in Display Subsystem. It stalls GE5 when appropriate and passes other handshakes through for accesses such as host to XMAP or host to RE.

The handshake control unit allows two types of transfers between the Host Interface Subsystem and the other MGR Subsystems. These two types of transfers are single word transfers and DMA transfers.

Stall Control Unit

Responsible for stalling the GE5 when a condition arises that requires it to be stalled. The full stall keeps the I/O Management unit in its current state and also keeps the PC, MEMPTR and REPTR from changing. The clock to the Weitek 3132 is fully suppressed and the entire GE5 is literally stalled.

PC Control Unit

Acts as a microcode sequencer for the GE5. The control unit controls microcode instruction execution sequencing through up to 32K words of microcode. The PC control unit allows branching and conditional jumps. It uses a 15 bit wide PC to access microcode code RAM. Microcode instructions are read from the address pointed to by the PC and are executed by the HQ1 and the Weitek 3132 chip.

MEMPTR Control Unit

Used as a data pointer for accessing words in microcode data RAM. MEMPTR is 14 bits wide allowing a maximum microcode data RAM size of 16K words. The MGR has only 8K words of microcode data RAM and each microcode data RAM word is 32 bits wide. The MEMPTR is loaded from microcode instructions to perform microcode data RAM accesses by the

microcode.

REPTR Control Unit

Used as an address pointer to RE registers. It is used by GE5 microcode to load RE registers. REPTR is not accessible by host which means that host cannot directly access RE registers. A microcode token GE_LOADRE allows the host to load RE registers with an assist from microcode. The host generally does not access RE register directly and exceptions will be discussed later.

GE Data Bus Buffer

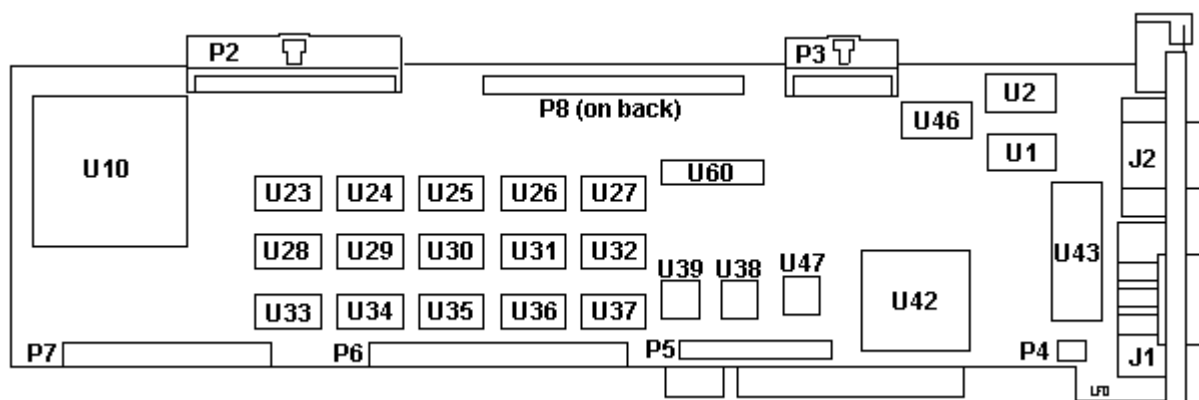
This is a 32 bit buffer which allows data to be passed between the GE5 bus and the host interface. Single word transfers use this data path to transfer data between the host and the microcode code RAM, the microcode data RAM and the HQ1 PC register. DMA transfers between the host and the microcode data RAM or between the host and the Raster Engine also use this data path.

Utility Bus Buffer

This is an eight bit transceiver which allows data to pass between the host bus and the utility bus. Only the low byte of the local bus are passed on to the utility bus. The address decoding and control signal generation are handled by the HQ1.

MRV2 Raster Video Card

FRU P/N 71F1151, P/N 71F1149   (photos from okqubit.net)



J1 3W3 video port

J2 HD15 GENLOCK port

P2 Geometry Engine Bus connector

P3 Utility Bus connector

P4 RGB Video connector

P5 Video Bus connector

P6 Utility Bus connector

P7 Pixel Bus connector

P8 Z Buffer Utility Bus connector

U1 30 MHz osc

U2 24.545452 MHz osc

U10 L1A4996 RE 2.1 C

U23-37 [TC524258AJ-10](#) 256Kx4 multiport DRAM

U38,39 [Bt431KJP](#) 64x64 Pixel Cursor Gen.

U42 XC2018-70

U43 27C256 ROM BIOS?

U46 111.518 MHz osc

U47 [Bt438KJP](#) Clock Gen. for RAMDAC

U60 MRV 1149 (VPD)

30 MHz osc - 15 MHz for PAL and SECAM

24.54542 MHz osc - 12.27 MHz for RS170 30 Hz (NTSC)

111.518 MHz osc - 1280x1024 at 60 Hz (NI) and 30 Hz (Int)

This is the Micro Channel Raster Video Engine. It is based upon the RE 2.1 raster engine chip from SGI. The raster engine provides all the per-fragment and -pixel operations. It also contains the raster scanning hardware and video signal generation circuitry. It only obtains power and ground from the bus, no other bus signals are used. In addition to the two ribbon cables from the Geometry Engine, a third connection services the Genlock features of the Raster Engine.

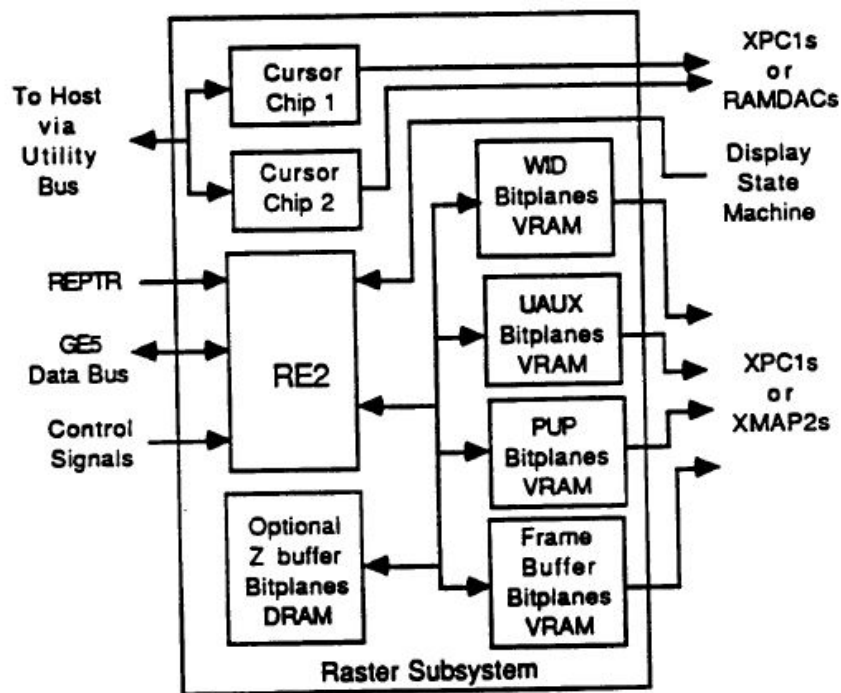


Figure 6.1 Raster Subsystem Block Diagram

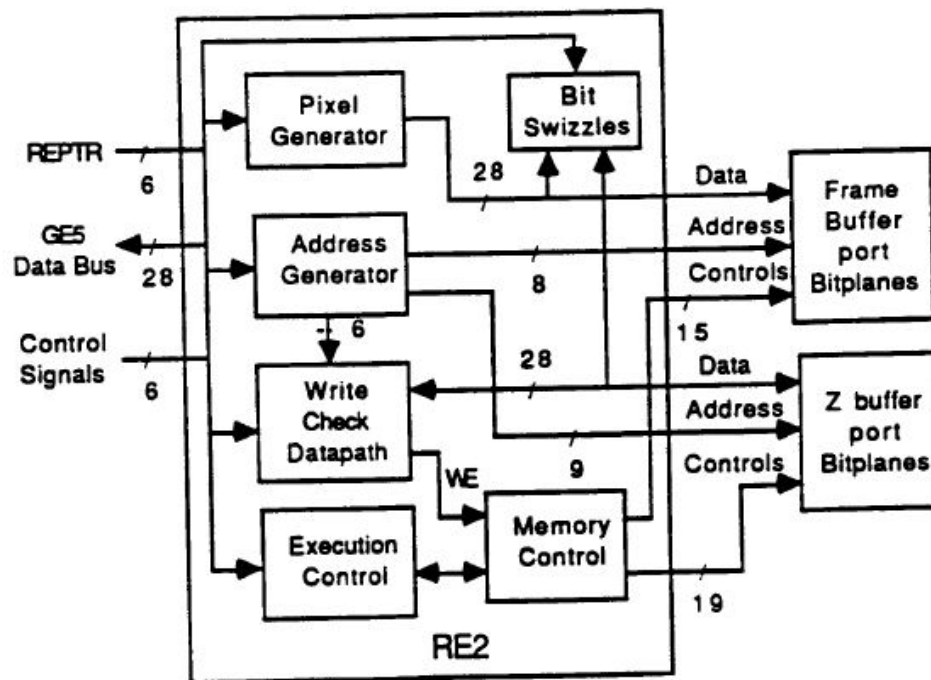


Figure 6.2 RE2 Block Diagram

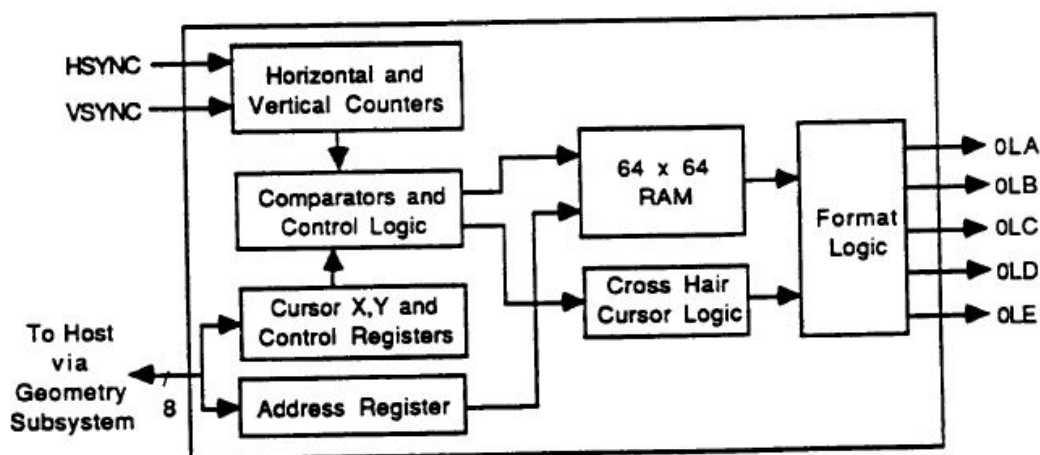


Figure 6.6 Cursor Chip Block Diagram

The Display State Machine (DSM) is built out of a XILINX 2018 programmable logic array, an 8K by 8-bit RAM, a Bt438 clock generator chip and other miscellaneous clock control circuitry. The 8K PROM contains the Xilinx logic configuration data and the four different monitor timing tables. On system reset the logic configuration is loaded into the Xilinx chip and the monitor timing tables are copied to the 2K RAM for faster access. The operation of the DSM is controlled by various bits in the display registers.



I cannot prove any of this... Right below U60 is two PLDs, a 16V8 and a 20V8.

Other than you can burn these to make anything you want, I've got nothing.

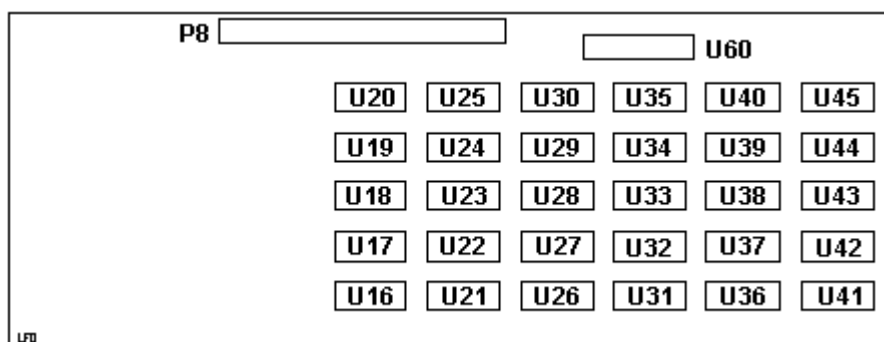
On the card edge connector, are the VGA passthrough connector and the HiRes video output connector. The Raster Video card contains the basic 8-bitplanes of framebuffer memory as well as 2 bitplanes of overlay framebuffer and 2 bits of window ID bitplanes, for a total of 12 bpp.

This card features an RGB video output connector (large w/plug-style connectors) and a genlock input/output connector (15-pin DSUB).

The MZB1 plugs onto the back (circuitboard side) of this card (use the short screws). The MEV2 plugs on to the front side (component side) of the card (use long screws).

MZB1 24-bit Z-Buffer Option

FRU P/N 42F6889, P/N 81F7923, ASSY # 030-8028-003 REV A  (photos from okqubit.net)



P8 Z-Buffer Utility Bus connector

U16-45 [TMS44C256DJ-10](#), [TC514256AJ-10](#) or compatible 256Kx4 DRAM

U60 MZB 7923 (VPD)

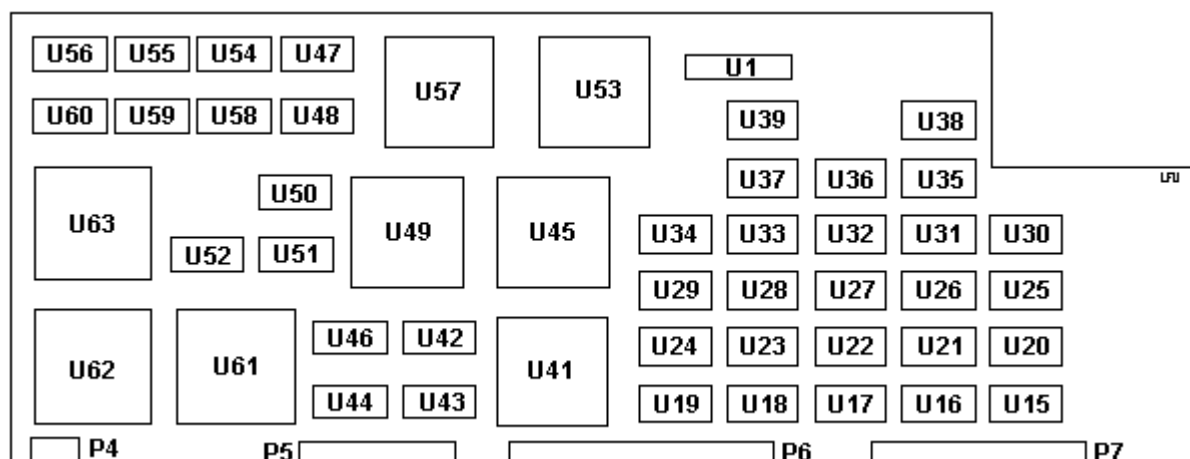
This is the optional 24-bit hardware Z-buffer card. The function of the Z-buffer is to perform hidden surface removal via a depth test mechanism. In the traditional application, each pixel's "Z" coordinate value is tested against the value already present in that location in the Z-buffer. If it is smaller (i.e. closer to the eye) the pixel is updated with the current value and the Z-buffer

is updated. By controlling the Z-buffer operation, a number of other useful operations can be performed.

For non-3D applications, the Z-buffer can be used as an off-screen memory buffer for saving the contents of the normal framebuffer. As no host to adapter memory transfer takes place, this operation is very fast. The Z-buffer is implemented in dynamic RAM (DRAM) and consists of 3.75 MB of DRAM.

MEV2 24-bit Frame Buffer Option

FRU P/N 71F1114, ASSY # 030-8027-002 REV A  (photos from okqubit.net)



P4 RGB Video connector

P5 Video Bus connector

P6 Utility Bus connector

P7 Pixel Bus connector

U1 MEV 1112 (VPD)

U15-39 [TC524258AJ-10](#) 256Kx4 multiport DRAM

U41,45,49,53,57 SGI XMAP2 L1A5081

U42-44,46-48,50-52,54-56,58-60 [IDT7164](#) 8Kx8 SRAM

U61-63 [Bt457KPJ125](#) 125 MHz 256 Color RAMDAC

This card supplies additional bitplanes of memory to bring the system to 24 bits per pixel of normal framebuffer in addition to 2 additional overlay/popup bitplanes and 2 additional window ID planes, for a total of 32-bits per pixel. A fully configured frame buffer has a total of 1280x1024*(32/8) or 5MB of video RAM (VRAM) implemented in 256Kx8 ICs.

You'll notice that the VRAM is laid out in a 5xN array of chips. Each chip supplies 256 horizontal pixels ($1280/5 = 256$). However, to achieve greater performance, the raster engine chip is designed to write up to 5 pixels at a time, so the five VRAM chips are interleaved; the first supplies pixels 1, 6, 11, etc. the second 2, 7, 12, etc. and so on.

The MEV2 card is a daughter board which attaches to the MRV2 card. It has a VPD PROM and provides an additional 20 bitplanes of VRAM which are part of the Raster subsystem.

It also contains the five XMAPP2 chips, the 8K color map chip and the three DAC chips which are part of the Display subsystem and are described in the chapter on the Display subsystem. The card has the Utility bus, Video bus, Pixel bus and RGB connections described above for the

MRV2 card.

ISA vs MCA MEV2

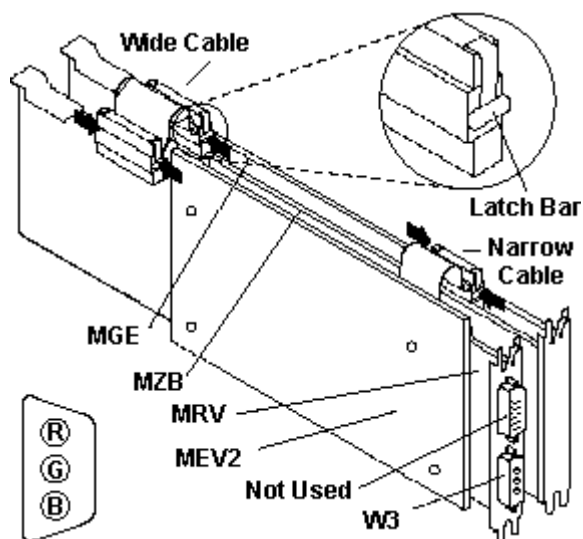
MEV2 ISA



MEV2 MCA



Of course, the internal traces may be different, but the PCB layouts sure look the same, except the ISA version lacks U1 [not needed for ISA].



Connector cable (wide) 53F3271 80 pin .025 pitch HPDB

Connector cable (narrow) 53F3272 40 pin .025 pitch HPDB

Removing Connector Cable

Squeeze the metal clips on the ends of the connectors and pull connector off header.

Plugging Connector Cables Back On

Squeeze the metal clips so that the loose ends of the clip enter the catch on the header.

The following block diagram is derived from the IrisVision Block Diagram on [The Very Unofficial IrisVision Home Page](#).

Video Timing Parameters

Video Resolution					
Parameter	High	Stereo	Medium	NTSC	PAL
Pixel Clock (MHz)	107.3	107.3	64.00	12.27	15.00
Horizontal Freq (kHz)	63.9	63.9	48.78	15.73	15.63
Frame Rate (Hz)	60.0	120.0	60.00	29.97	25.00
Field Rate (Hz)	60.0	120.0	60.00	59.94	50.00
Horizontal					
Visible Pixels	1280	1280	1024	640	768
Line Period (us)	15.70		20.63	63.56	64.00
Blanking (us)	3.73		4.69	11.00	12.00
Front Porch (us)	0.28		1.10	1.63	1.67
Sync Width (us)	1.12		1.25	4.89	4.67
Back Porch (us)	2.34		2.34	4.48	5.67
Vertical					
Visible Lines	1024	492	768	480	576
Front Porch (ms)	0.047		0.062	0.191	0.160
Sync Width (ms)	0.047		0.062	0.191	0.160
Back Porch (ms)	0.450		0.804	0.890	1.280

Original TXT version mirrored [HERE](#).

ADF Sections AdapterId 8EE6h "SGI Micro-Channel IRISVISION Adapter"

Interrupt Level

Determines the interrupt level used by IRISVISION Adapter.

<"Level 2 ">, 3, 4, 5, 6, 9, 10, 11, 12

Memory Mapped I/O Address Range

Determines the range of Memory Mapped I/O addresses used by the IRISVISION Adapter. The addresses in this range cannot be used by any other installed device.

<"0C0000 to 0C7FFF">, 0C8000 to 0CFFFF, 0D0000 to 0D7FFF, 0D8000 to 0DFFFF

Arbitration Level

Determines the bus arbitration level used.

[<" Arb Level 1 ">](#), 0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14

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Ardent Tool of Capitalism is maintained by Tomáš Slavotínek.

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